

PRODUCT AND CATEGORY MANAGEMENT

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	INT	Primary Key
Category_Name	VARCHAR2(50)	NOT NULL UNIQUE
Description	VARCHAR2(200)	-

1.2 Product Table

Attribute	Data Type	Constraint
Product_ID	INT	Primary Key
Product_Name	VARCHAR2(100)	NOT NULL
Category_ID	INT	Foreign Key
Brand	VARCHAR2(50)	—
Price	NUMBER(10,2)	NOT NULL
Stock	INT	NOT NULL

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: One-to-Many (1:M)

3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID INT PRIMARY KEY,  
3     Category_Name VARCHAR(50) UNIQUE NOT NULL,  
4     Description VARCHAR(200)  
5 );
```

Table created.

3.2 Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID INT PRIMARY KEY,  
3     Product_Name VARCHAR2(100) NOT NULL,  
4     Category_ID INT,  
5     Brand VARCHAR2(50),  
6     Price NUMBER(10,2),  
7     Stock INT NOT NULL,  
8     CONSTRAINT fk_category  
9         FOREIGN KEY (Category_ID)  
10        REFERENCES Category(Category_ID)  
11 );
```

Table created.

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category VALUES
  2  (101, 'Men Watches', 'Stylish watches for men');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2  (102, 'Women Watches', 'Elegant watches for women');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2  (103, 'Smart Watches', 'Digital and smart wearable watches');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2  (104, 'Luxury Watches', 'Premium luxury watch collection');

1 row created.

SQL>
SQL> COMMIT;
```

4.2 Product Records

```
SQL> INSERT INTO Product
  2  VALUES (201, 'Titan Neo', 101, 'Titan', 4999, 25);

1 row created.

SQL> INSERT INTO Product
  2  VALUES (202, 'Fastrack Reflex', 103, 'Fastrack', 2999, 40);

1 row created.

SQL> INSERT INTO Product
  2  VALUES (203, 'Casio Vintage', 102, 'Casio', 3499, 18);

1 row created.

SQL> INSERT INTO Product
  2  VALUES (204, 'Fossil Grant', 104, 'Fossil', 8999, 12);

1 row created.

SQL> INSERT INTO Product
  2  VALUES (205, 'Sonata Classic', 101, 'Sonata', 1999, 30);

1 row created.

SQL> COMMIT;
Commit complete.
```

5. Data Retrieval

5.1 Display Category Records

```

SQL> SELECT * FROM Category;

CATEGORY_ID CATEGORY_NAME
-----
DESCRIPTION
-----
          101 Men Watches
Stylish watches for men

          102 Women Watches
Elegant watches for women

          103 Smart Watches
Digital and smart wearable watches

CATEGORY_ID CATEGORY_NAME
-----
DESCRIPTION
-----
          104 Luxury Watches
Premium luxury watch collection

```

5.2 Display Product Records

```
SQL> SELECT * FROM Product;
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
CATEGORY_ID BRAND
```

```
PRICE
```

```
STOCK
```

```
201
```

```
Titan Neo
```

```
101 Titan
```

```
4999
```

```
25
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
CATEGORY_ID BRAND
```

```
PRICE
```

```
STOCK
```

```
202
```

```
Fastrack Reflex
```

```
103 Fastrack
```

```
2999
```

```
40
```

6. Product Update

```
SQL> UPDATE Product
  2 SET Price = 5499
  3 WHERE Product_ID = 201;
```

```
1 row updated.
```

7. Product Deletion

```
SQL> DELETE FROM Product
2 WHERE Product_ID = 205;

1 row deleted.
```

8. Category-wise Product Reports

```
SQL> SELECT
2     Category_ID,
3     COUNT(*) AS Total_Products,
4     SUM(Price) AS Total_Price,
5     AVG(Price) AS Average_Price
6 FROM Product
7 GROUP BY Category_ID
8 ORDER BY Category_ID;
```

CATEGORY_ID	TOTAL_PRODUCTS	TOTAL_PRICE	AVERAGE_PRICE
101	1	5499	5499
102	1	3499	3499
103	1	2999	2999
104	1	8999	8999

8.1 Product Count by Category

```
SQL> SELECT
2     Category_ID,
3     COUNT(Product_ID) AS Total_Products
4 FROM Product
5 GROUP BY Category_ID
6 ORDER BY Category_ID;
```

CATEGORY_ID	TOTAL_PRODUCTS
101	1
102	1
103	1
104	1